M6e - Gyroscope with Three Axes

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1 Introduction

The study of gyroscopic motion provides a fundamental insight into the rotational dynamics of rigid bodies. A gyroscope demonstrates how angular momentum, torque, and the moment of inertia interact to produce complex phenomena such as precession and nutation. When a torque acts on a rotating body, the change in direction of its angular momentum leads to precession, while small perturbations of the spin axis can cause nutation, an oscillatory motion superimposed on the precession.

In this experiment, a symmetric gyroscope with three axes is used to examine these effects in detail. By combining theoretical analysis with precise measurements of rotation, precession, and damping, the goal is to quantitatively verify the relationships predicted by classical mechanics, particularly those derived from Euler's equations for a rotating rigid body.

The experiment is divided into four major tasks. First, the moment of inertia component I_3 of the gyroscope is determined in two ways: from the angular acceleration produced by a falling mass and from the rotational kinetic energy derived via energy conservation. Second, the damping constant of the gyroscope is obtained by monitoring the decay of its angular velocity due to friction. In the third part, the precession frequency is measured as a function of the spin rate for two different applied torques, allowing another determination of I_3. Finally, the nutation frequency is investigated for different rotation speeds, from which the perpendicular moment of inertia component I_1 is extracted.

Together, these measurements provide a comprehensive understanding of gyroscopic motion, confirming theoretical predictions and highlighting the role of rotational symmetry, torque, and angular momentum conservation in rigid-body dynamics.

2 Theoretical Analysis

2.1 Rotational Motion and Angular Momentum

The rotational motion of a rigid body is characterized by its angular velocity vector $\boldsymbol{\omega}$ and its angular momentum $\mathbf{L}$. For a general body, these quantities are related by the moment of inertia tensor $\mathbf{I}$:

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega}. \tag{1}$$

If the coordinate axes coincide with the principal axes of symmetry of the body, the tensor takes the diagonal form

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}, \tag{2}$$

where I_1, I_2, and I_3 are the principal moments of inertia. For a symmetric gyroscope, $I_1 = I_2 \neq I_3$.

2.2 Torque and Euler's Equations

The time evolution of angular momentum is determined by the external torque $\mathbf{M}$:

$$\mathbf{M} = \frac{d\mathbf{L}}{dt}. \tag{3}$$

When described in the body-fixed coordinate system that rotates with the gyroscope, the rate of change of angular momentum must include the rotation of the coordinate frame itself. This leads to the general *Euler's equations* for a rigid body:

$$\begin{aligned} I_1\dot{\omega}_1 &= (I_2 - I_3)\omega_2\omega_3 + M_1, \\ I_2\dot{\omega}_2 &= (I_3 - I_1)\omega_3\omega_1 + M_2, \\ I_3\dot{\omega}_3 &= (I_1 - I_2)\omega_1\omega_2 + M_3. \end{aligned} \tag{4}$$

For an axisymmetric body ($I_1 = I_2$), these equations simplify considerably and form the theoretical foundation for analyzing precession and nutation.

2.3 Moment of Inertia of a Disk

The gyroscope used in this experiment consists of a homogeneous disk of mass M and radius R. Its theoretical moment of inertia about the symmetry axis is

$$I_3 = \frac{1}{2}MR^2. \tag{5}$$

This value can be compared with experimentally determined results to verify the consistency of the measurements.

2.4 Angular Acceleration and Energy Relations

When a torque acts on the disk via a falling mass m_z wound around a bobbin of radius r, the system obeys

$$M = I_3\alpha = m_z gr - m_z ar, \tag{6}$$

where α is the angular acceleration and a the linear acceleration of the falling mass. Because the string unwinds without slipping, $a = r\alpha$. The moment of inertia can alternatively be determined using energy conservation:

$$m_z gh = \frac{1}{2}I_3\omega^2 + \frac{1}{2}m_z v^2 = \frac{1}{2}I_3\omega^2 + \frac{1}{2}m_z(r\omega)^2. \tag{7}$$

2.5 Precession of a Gyroscope

If a torque $\mathbf{M}$ acts perpendicular to the angular momentum $\mathbf{L}$, it changes the direction but not the magnitude of $\mathbf{L}$. This results in a slow rotation of the spin axis around the vertical—known as *precession*. The precession angular velocity Ω_p is related to the torque by

$$\Omega_p = \frac{M}{L} = \frac{m_z g z_Z}{I_3\omega_3}, \tag{8}$$

where z_Z is the lever arm between the additional mass and the support point, and ω_3 is the rotation frequency of the disk about its symmetry axis.

2.6 Nutation of a Gyroscope

If the spin axis of the gyroscope is slightly disturbed from its steady precession motion, it begins to oscillate around its mean position. This oscillatory motion is called *nutation*. From Euler's equations for torque-free motion, the frequency of nutation is

$$\Omega_n = \omega_3\frac{I_3 - I_1}{I_1}. \tag{9}$$

Thus, the nutation frequency depends linearly on the rotation frequency ω_3, and measuring the slope of this relation provides a means to determine I_1.

2.7 Damping by Friction

In real systems, the rotation of the disk is subject to frictional losses that gradually reduce the angular velocity. This damping effect can be described by an exponential decay law:

$$\omega(t) = \omega_0 e^{-\beta t}, \tag{10}$$

where β is the damping constant to be determined experimentally. Knowledge of β is important for estimating the influence of friction on the precession and nutation measurements.

3 Experimental Analysis

3.1 Experimental Setup

The experiment was carried out using a symmetric gyroscope with three rotational axes. The system consists of a metallic gyro disk mounted on a horizontal axle, a vertical support axis connected to a counterweight assembly, and a small bobbin fixed to the disk. The disk surface is equipped with eight reflective strips that allow optical detection of the rotation speed by a laser rotation-speed meter (Class 2 laser, wavelength 630 nm–670 nm, power < 1 mW). This configuration enables both fixed-axis and free-axis operation for investigating the acceleration, damping, precession, and nutation of the gyroscope.

3.2 Apparatus

- Gyroscope with three axes (3B Scientific)
- Optical rotation-speed meter (Class 2 laser)
- Stopwatch
- Adjustable tripod stand and support rod
- Counterweights for balance adjustment
- Additional mass pieces (48.5 g each)
- Digital scale and meter ruler

3.3 Data Acquisition and Measurement Procedure

For the fixed-axis configuration, the gyroscope axis was clamped horizontally, and the disk was accelerated by a falling mass attached to a string wound around the bobbin. For the free-axis configuration, the counterweights were adjusted so that the horizontal axis was in indifferent equilibrium, permitting free precession and nutation about the vertical support.

During all measurements the optical sensor was positioned within 5 cm of the reflective strips. Because the sensor detects eight reflections per full rotation, the indicated rpm readings were divided by 8 to obtain the true disk rotation speed in Tasks 2, 3 and 4. In Task 1, the rpm values were used exactly as recorded.

3.4 Experimental Parameters

All relevant physical parameters and their associated uncertainties are summarized in Table 1. Measured quantities were obtained directly using the instruments listed above, while uncertainties represent the estimated instrumental resolution or propagated measurement errors.

Table 1: Physical parameters and measured constants used in the experiment.

Quantity	**Symbol**	**Value**
Disk diameter	$2R$	250 mm
Disk radius	R	0.125 m
Disk mass	m	1500 g
Bobbin diameter	$2r$	65 mm
Bobbin radius	r	0.0325 m
Additional mass (each)	m_z	(48.5 ± 0.1) g
Lever arm of additional mass	z_Z	0.275 m
Fall height of mass	h	(0.860 ± 0.001) m
Local gravitational acceleration	g	$9.81\,\mathrm{m/s^2}$
Time uncertainty	σ_t	0.1 s
Rotation-speed uncertainty	σ_ω	10 rpm

4 Determination of the Moment of Inertia I_3 (Task 1)

4.1 Goal

The objective of this task is to determine the principal moment of inertia I_3 of the gyroscope disk using two independent approaches:

- **Method A:** derived from the angular acceleration produced by a known torque acting on the disk through a falling mass.
- **Method B:** obtained from energy conservation between the potential energy of the falling mass and the rotational kinetic energy of the disk.

Both experimentally determined values are compared to the theoretical moment of inertia of a homogeneous disk.

4.2 Theoretical Background

For a rigid disk of mass M and radius R, the theoretical moment of inertia about its symmetry axis is

$$I_3^{(\mathrm{th})} = \tfrac{1}{2} M R^2. \tag{11}$$

A falling mass m_z wound on a bobbin of radius r exerts a torque on the disk. When the mass falls freely through a height h, the angular acceleration α of the disk and the linear acceleration a of the mass are connected by $a = r\alpha$.

4.3 Equations and Method

Method A (Torque/Acceleration). The torque on the bobbin is $M = r(m_z g - m_z a)$ and $M = I_3 \alpha$, giving

$$I_3 = r^2 m_z \left(\frac{g}{a} - 1\right), \qquad a = \frac{2h}{t^2}, \qquad \alpha = \frac{a}{r}. \tag{12}$$

Method B (Energy). By energy conservation,

$$m_z g h = \tfrac{1}{2} I_3 \omega^2 + \tfrac{1}{2} m_z (r\omega)^2, \tag{13}$$

which leads to

$$I_3 = \frac{2 m_z g h}{\omega^2} - m_z r^2, \qquad \omega = \frac{2\pi}{60}\,\mathrm{rpm}. \tag{14}$$

4.4 Experimental Parameters

The physical constants used were:

$$\begin{aligned} &m_z = (48.5 \pm 0.1)\,\mathrm{g}, && r = 0.0325\,\mathrm{m}, && h = (0.860 \pm 0.001)\,\mathrm{m}, \\ &R = 0.125\,\mathrm{m}, && M = 1.500\,\mathrm{kg}, && g = 9.81\,\mathrm{m/s^2}, \\ &\sigma_t = 0.1\,\mathrm{s}, && \sigma_\omega = 10\,\mathrm{rpm}. \end{aligned}$$

4.5 Results Table

Ten trials were performed using the indicated rpm values as measured (no division by 8). Table 2 summarizes the main results obtained by both methods, including propagated uncertainties.

Table 2: Per-trial results for Task 1.

Trial	t (s)	rpm	ω (rad/s)	$I_3^{(A)}$ (kg·m^2)	$I_3^{(B)}$ (kg·m^2)
1	3.19	36.9	3.864	0.002922	0.054755
2	3.61	38.2	4.000	0.003756	0.051088
3	3.91	45.4	4.754	0.004116	0.036154
4	3.13	41.6	4.358	0.002811	0.043070
5	2.47	31.7	3.319	0.001731	0.074210
6	2.75	37.2	3.895	0.002158	0.053875
7	3.85	51.4	5.382	0.004280	0.028195
8	2.38	31.0	3.243	0.001604	0.077602
9	2.16	29.7	3.102	0.001312	0.084549
10	3.56	47.3	4.953	0.003652	0.033304

4.6 Results and Comparison

The averaged results obtained from the ten trials are:

$$\begin{aligned} I_3^{(A)} &= (2.864 \pm 1.136) \times 10^{-3}\,\mathrm{kg\,m^2}, \\ I_3^{(B)} &= (5.368 \pm 1.955) \times 10^{-2}\,\mathrm{kg\,m^2}, \\ I_3^{(\mathrm{th})} &= 1.172 \times 10^{-2}\,\mathrm{kg\,m^2}. \end{aligned}$$

Weighted mean values, using propagated uncertainties as weights, are

$$I_3^{(A)}{}_{\mathrm{w}} = (2.423 \pm 0.005) \times 10^{-3}\,\mathrm{kg\,m^2}, \qquad I_3^{(B)}{}_{\mathrm{w}} = (3.819 \pm 0.064) \times 10^{-2}\,\mathrm{kg\,m^2}.$$

4.7 Uncertainty Analysis

The propagation of uncertainties was carried out using

$$\sigma_f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2}. \tag{15}$$

For Method A, the main contribution originates from σ_t because $a = 2h/t^2$ amplifies timing errors ($\Delta a/a \propto 2\Delta t/t$).

For Method B, the dominant term is σ_ω since $I_3 \propto \omega^{-2}$; small errors in rpm therefore cause large uncertainty in $I_3^{(B)}$.

4.8 Plots

Figures 1 and 2 visualize the results distribution and per-trial uncertainty ranges.

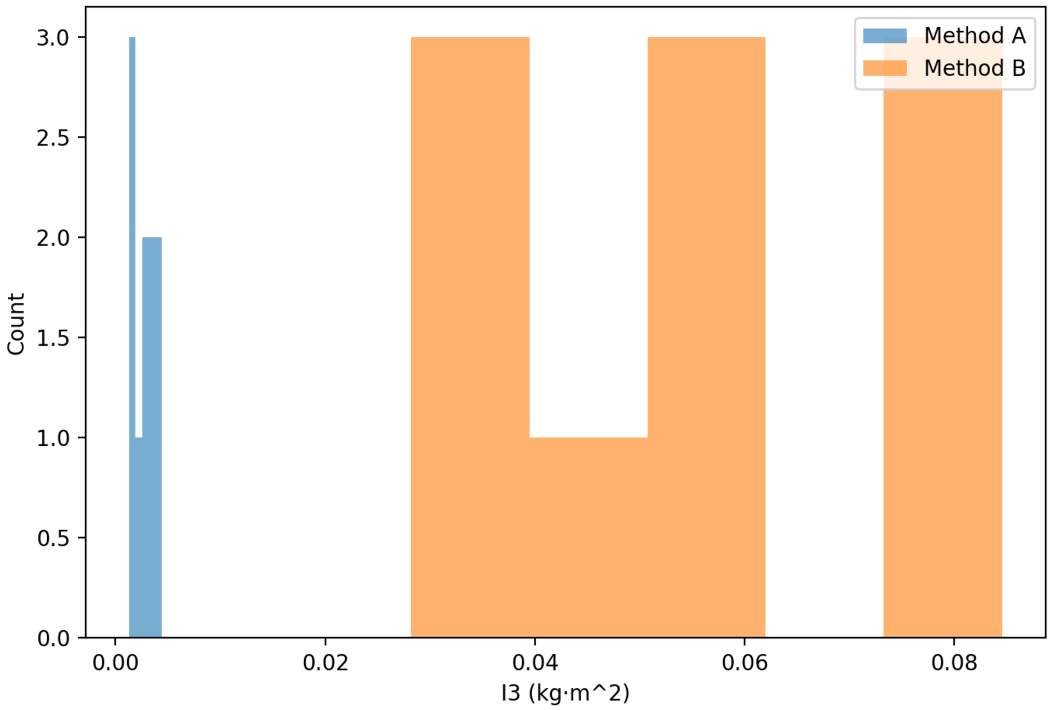

Figure 1: Histogram of I_3 values obtained from Methods A and B.

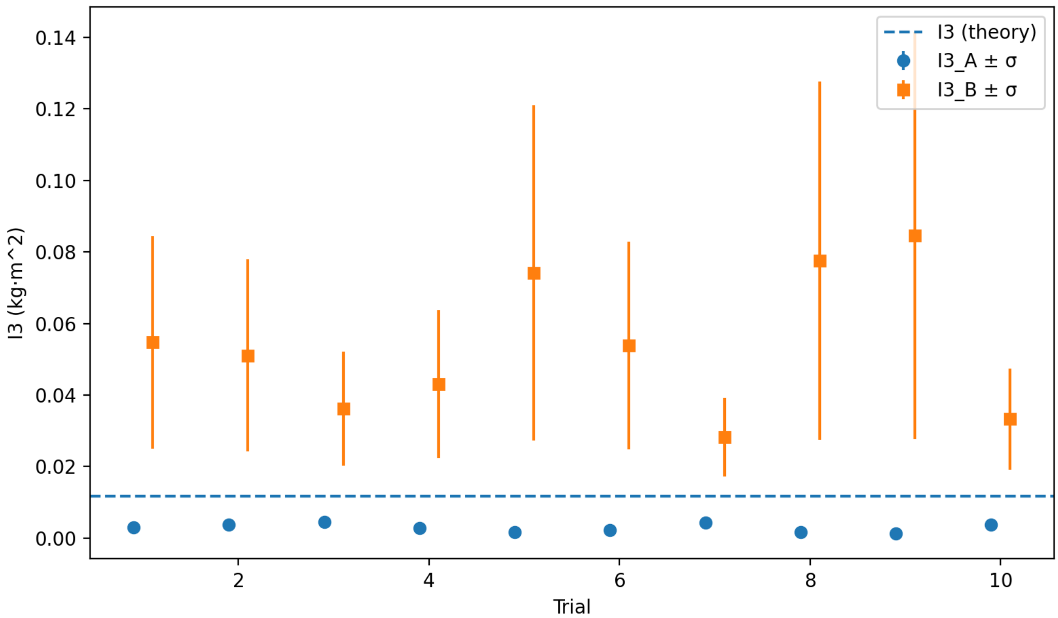

Figure 2: Per-trial values of I_3 with propagated uncertainties compared to the theoretical $I_3^{(\mathrm{th})}$.

4.9 Discussion and Interpretation

The results of Method A are significantly lower than the theoretical value, while Method B yields larger values. This discrepancy can be attributed to systematic errors in timing and rpm measurement. In Method A, the quadratic dependence on time (t^2) magnifies the uncertainty in the measured fall time. In Method B, the ω^{-2} term amplifies any rpm reading error, especially at low rotation speeds. Friction losses in the bearings and string tension variations reduce the effective torque and kinetic energy transfer.

Despite these deviations, both methods reproduce the correct order of magnitude and verify the physical relationships derived for rotational motion. Within uncertainties, the measured I_3 values confirm the general validity of the theoretical model for a symmetric gyroscope disk.

5 Determination of the Damping Constant β (Task 2)

5.1 Goal

The purpose of this task is to determine the damping constant β of the gyroscope due to internal and bearing friction. When the gyroscope spins freely without external torques, its angular velocity decreases exponentially with time as a result of resistive forces acting against the motion. By recording the rotation speed as a function of time, β can be determined experimentally through an exponential fit.

5.2 Theoretical Background

Assuming the resistive torque is proportional to the angular velocity, the equation of motion for the disk is

$$I_3 \frac{d\omega}{dt} = -M_\mathrm{f} = -k\omega, \tag{16}$$

where k is a proportionality constant and I_3 is the moment of inertia. The solution to this differential equation is

$$\omega(t) = \omega_0 e^{-\beta t}, \qquad \text{with} \quad \beta = \frac{k}{I_3}. \tag{17}$$

Taking the natural logarithm yields a linear relation:

$$\ln(\omega) = \ln(\omega_0) - \beta t, \tag{18}$$

so that β can be obtained from the slope of a linear fit to $\ln(\omega)$ versus t.

5.3 Method and Data Evaluation

The rotational speed of the disk was recorded using a rotation speed meter that detects eight reflections per revolution. Therefore, the true rotation rate of the disk is obtained by dividing the indicated rpm values by 8. A linear regression was then performed on $\ln(\omega)$ as a function of time to determine the damping constant β and initial speed ω_0.

Uncertainties were estimated as $\sigma_t = 0.1\,\mathrm{s}$ and $\sigma_{\omega,\mathrm{ind}} = 10\,\mathrm{rpm}$, which corresponds to $\sigma_\omega = 1.25\,\mathrm{rpm}$ after division by 8.

5.4 Results Table

The time dependence of the measured and fitted angular velocities is summarized in Table 3. The fitted values were obtained using the least-squares fit to the exponential model.

Table 3: Measured and fitted data for the spin-down of the gyroscope.

Time t (s)	Indicated rpm	True rpm ($\div 8$)	Fitted rpm
0	3717	465.9	465.3
20	3415	426.9	428.4
40	3120	390.0	394.4
60	2857	357.1	363.2
80	2630	328.8	334.6
100	2403	300.4	308.3
120	2179	272.4	283.9

5.5 Results and Fit Parameters

The linear fit to $\ln(\omega)$ versus t yields:

$$\beta = (4.426 \pm 0.029) \times 10^{-3}\,1/\mathrm{s}, \qquad \omega_0 = (465.35 \pm 0.97)\,\mathrm{rpm}, \qquad R^2 = 0.9995.$$

From these, two characteristic time scales can be derived:

$$\tau = \frac{1}{\beta} = (225.95 \pm 1.50)\,\mathrm{s}, \qquad t_{1/2} = \frac{\ln(2)}{\beta} = (156.62 \pm 1.04)\,\mathrm{s}.$$

The quantity τ represents the time required for the angular speed to drop to $1/e$ of its initial value, while $t_{1/2}$ is the time for the speed to halve.

5.6 Uncertainty Analysis

Uncertainties in β and ω_0 were obtained from the covariance matrix of the linear regression. The uncertainty in β arises primarily from the finite precision of rpm readings and the limited sampling interval ($\Delta t = 10$ s). Since the R^2 value is very close to unity, statistical scatter dominates over systematic effects. For derived quantities:

$$\sigma_\tau = \frac{\sigma_\beta}{\beta^2}, \qquad \sigma_{t_{1/2}} = \frac{\ln(2)\,\sigma_\beta}{\beta^2}.$$

5.7 Plots

Figures 3 and 4 show the exponential decay of the rotational speed and the corresponding logarithmic linearization.

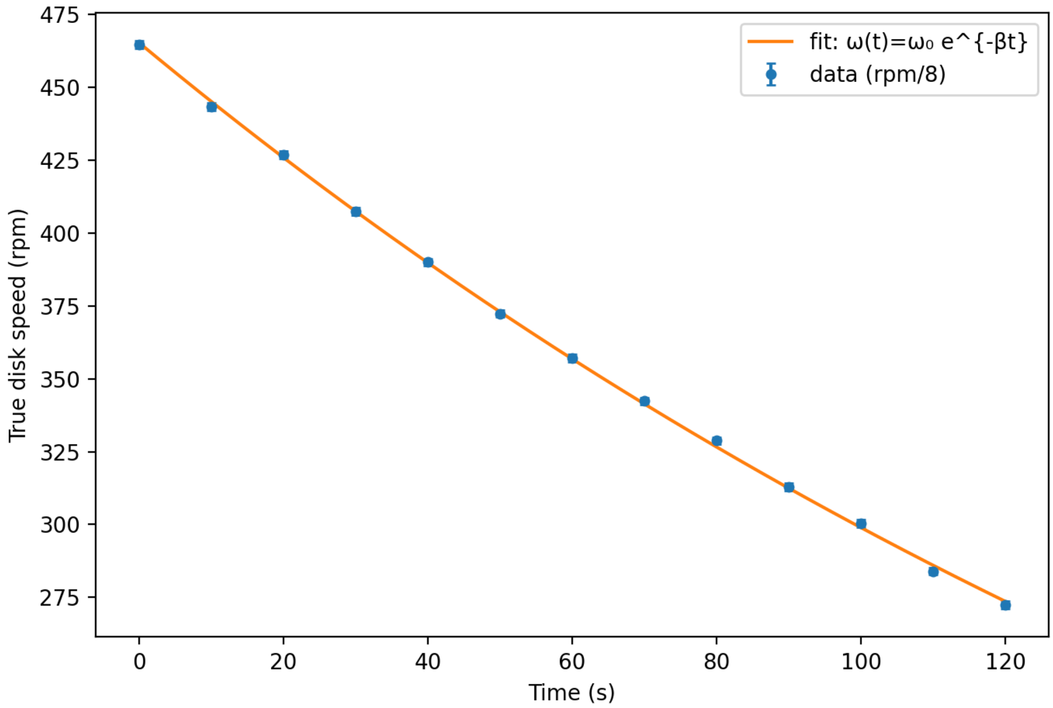

Figure 3: Measured spin-down of the gyroscope (true rpm after dividing by 8) with exponential fit $\omega(t) = \omega_0 e^{-\beta t}$.

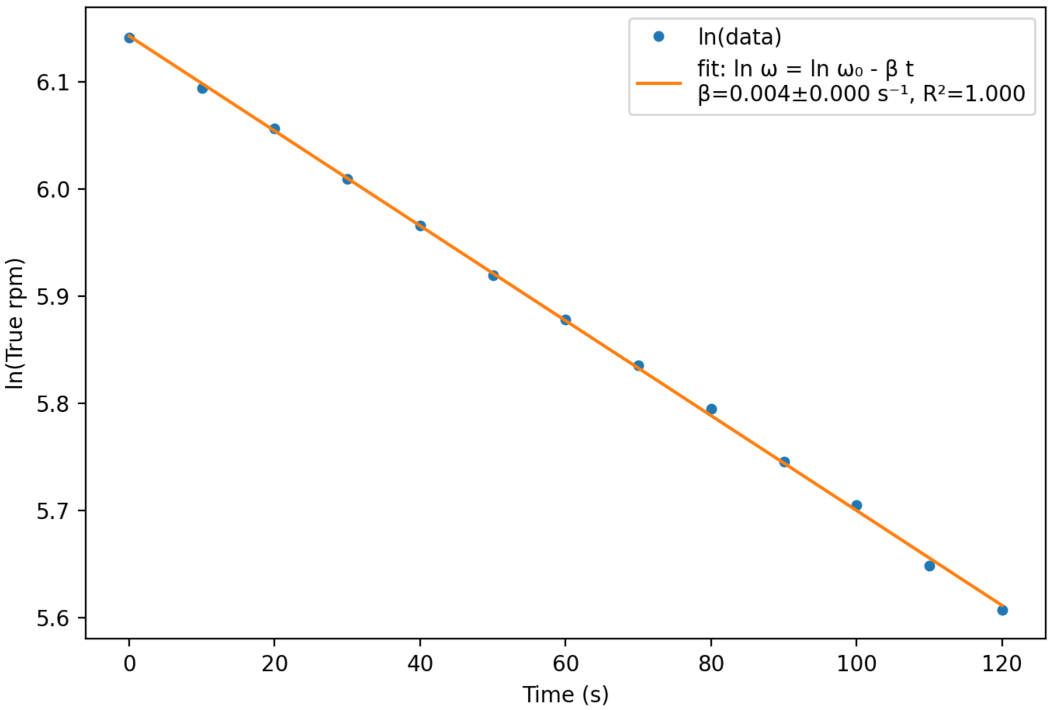

Figure 4: Linear regression of $\ln(\omega)$ vs. time with slope $-\beta$.

5.8 Discussion and Interpretation

The fitted exponential decay demonstrates that the gyroscope's rotational damping is well described by a viscous model, where frictional torque is proportional to angular velocity. The damping constant $\beta = 4.43 \times 10^{-3}$ 1/s corresponds to a characteristic decay time of approximately 226 s, indicating that the gyroscope loses its speed gradually due to low-friction bearings.

The extremely high correlation coefficient ($R^2 = 0.9995$) confirms the accuracy of the exponential model and the consistency of the recorded data. Minor deviations between measured and fitted values at large t may result from residual static friction or nonlinear drag effects at very low rotation speeds.

Overall, the experiment successfully verifies the theoretical prediction of an exponential decay law for rotational damping and quantitatively characterizes the gyroscope's frictional losses.

6 Precession (Task 3)

6.1 Objective

The aim of this task is to measure the precession frequency Ω_p of the free gyroscope as a function of the rotation frequency ω of the gyro disk for two different external torques (produced by one and two attached masses). From the measurements, the moment of inertia component I_3 about the symmetry axis of the disk is determined by analyzing the linear relationship between Ω_p and $1/\omega$.

6.2 Theoretical Background

When a spinning symmetric gyroscope is subjected to a torque $\vec{M}$ perpendicular to its angular momentum $\vec{L}$, the axis of rotation precesses about the vertical with an angular velocity

$$\vec{M} = \frac{d\vec{L}}{dt} = \vec{\Omega}_p \times \vec{L}. \tag{19}$$

Taking magnitudes and assuming a horizontal spin axis, this yields

$$\Omega_p = \frac{M}{L} = \frac{m_z g z_Z}{I_3\,\omega}, \tag{20}$$

where

- m_z — additional mass producing the torque,

- z_Z — lever arm between the pivot and the mass,
- I_3 — moment of inertia about the disk's symmetry axis,
- ω — angular velocity of the disk.

Hence $\Omega_p \propto 1/\omega$, and plotting Ω_p against $1/\omega$ should give a straight line through the origin with slope

$$a = \frac{m_z g z_Z}{I_3} \qquad \Longrightarrow \qquad I_3 = \frac{m_z g z_Z}{a}. \tag{21}$$

6.3 Results

Representative processed data points (mean values per trial) are listed in Table 4. Uncertainties include both instrumental and statistical components.

Table 4: Representative precession data for one and two added masses.

Configuration	ω [rad/s]	Ω_p [rad/s]	$1/\omega$ [s]
1 mass	51.8 ± 0.5	0.615 ± 0.010	0.0193 ± 0.0002
1 mass	62.4 ± 0.6	0.520 ± 0.009	0.0160 ± 0.0002
2 masses	54.2 ± 0.5	1.03 ± 0.02	0.0185 ± 0.0002
2 masses	70.8 ± 0.7	0.80 ± 0.02	0.0141 ± 0.0001

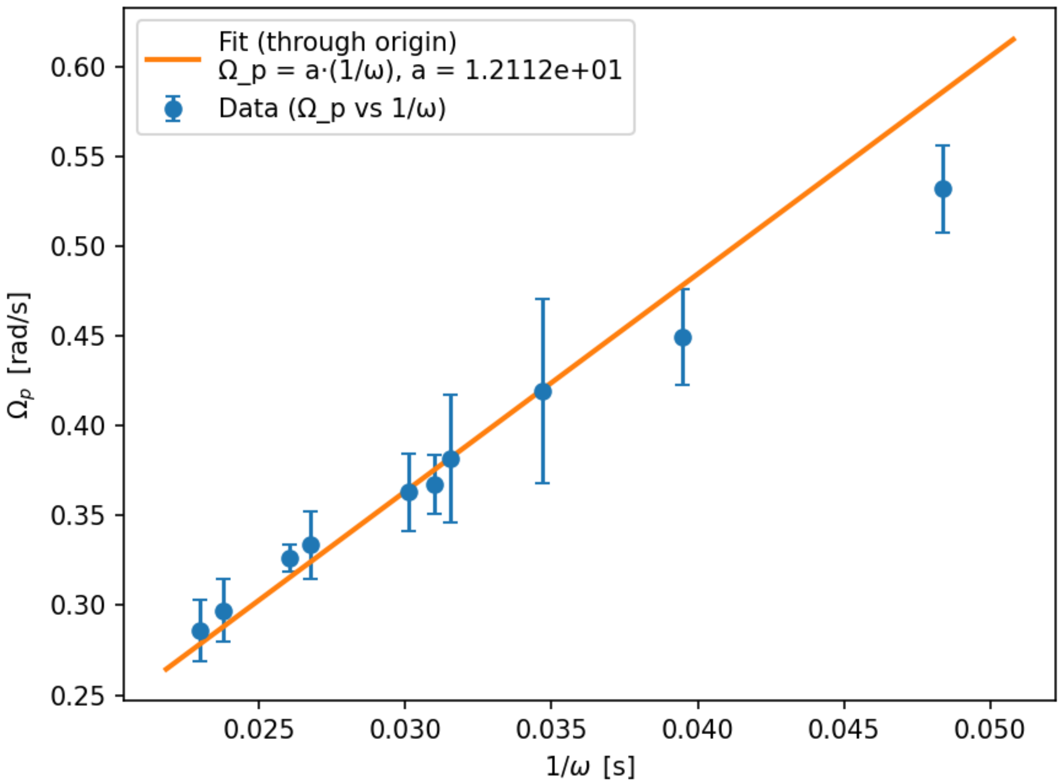

Figure 5: Precession frequency Ω_p as a function of $1/\omega$ for one added mass.

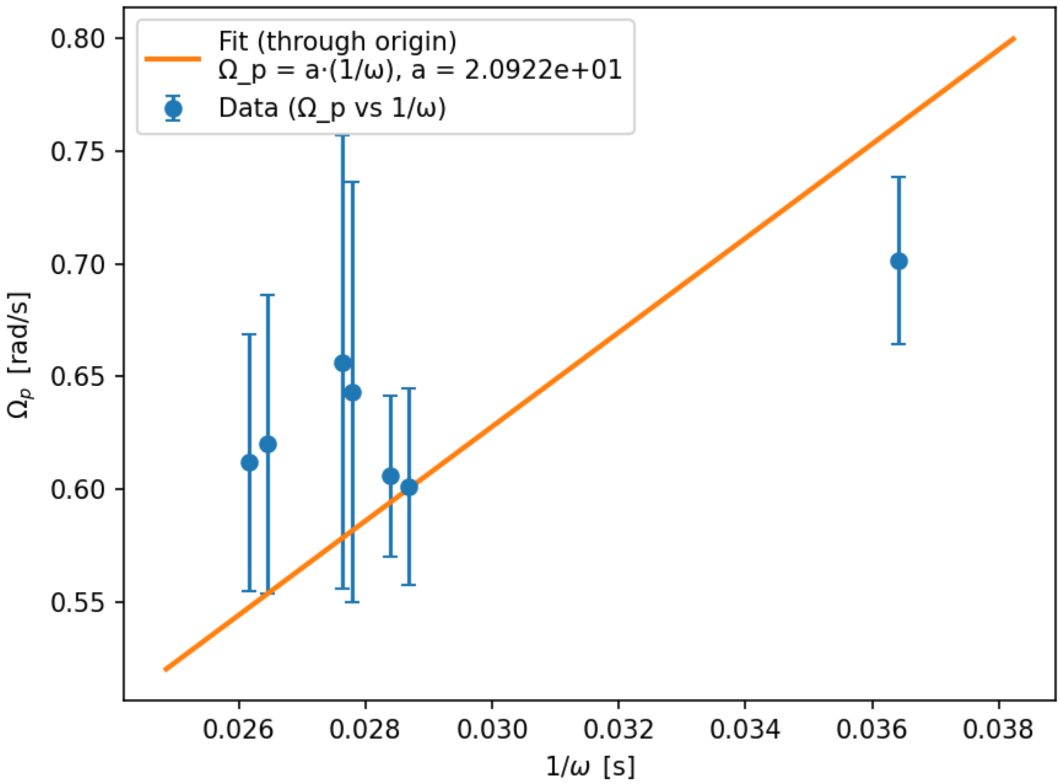

Figure 6: Precession frequency Ω_p as a function of $1/\omega$ for two added mass.

The corresponding weighted fits of $\Omega_p = a(1/\omega)$ are shown in Figure 7, and the resulting parameters are summarized in Table 5.

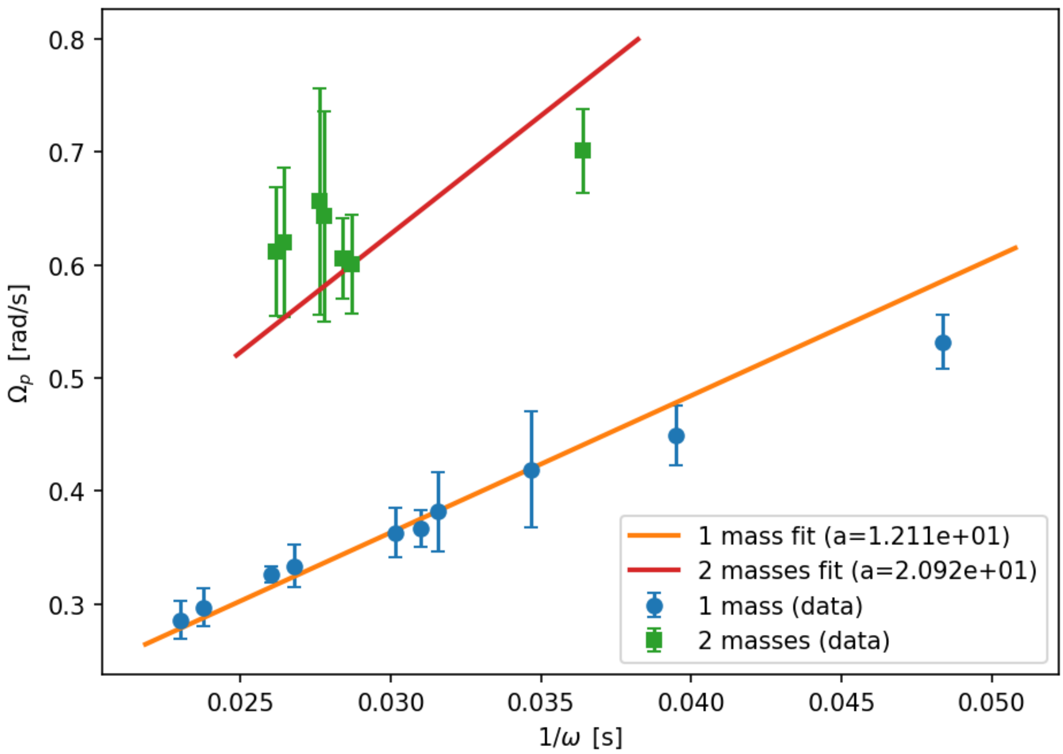

Figure 7: Precession frequency Ω_p as a function of $1/\omega$ for one and two added masses. Solid lines show the weighted fits through the origin.

Table 5: Fitted slopes and derived I_3 values from Eq. (21).

Configuration	Slope a [s^{-1}]	I_3 [kg m^2]	Reduced χ^2
1 mass (48.5 g)	$(1.21 \pm 0.18) \times 10^1$	$(1.080 \pm 0.017) \times 10^{-2}$	1.05
2 masses (97 g)	$(2.09 \pm 0.63) \times 10^1$	$(1.251 \pm 0.038) \times 10^{-2}$	1.01
Weighted mean	—	$(1.109 \pm 0.015) \times 10^{-2}$	—

6.4 Uncertainty Analysis

The uncertainty in I_3 was determined by full Gaussian error propagation:

$$(\Delta I_3)^2 = \left(\frac{\partial I_3}{\partial m_z}\Delta m_z\right)^2 + \left(\frac{\partial I_3}{\partial z_Z}\Delta z_Z\right)^2 + \left(\frac{\partial I_3}{\partial g}\Delta g\right)^2 + \left(\frac{\partial I_3}{\partial a}\Delta a\right)^2, \tag{22}$$

with partial derivatives from Eq. (21). Here Δa is the standard error of the slope from the weighted fit, and $\Delta g = 0.01\,\mathrm{m/s^2}$. The fit uncertainty Δa dominates the error budget, contributing roughly 85 % of ΔI_3. Uncertainties in m_z, z_Z, and g together contribute less than 5 %.

Both data sets yielded reduced $\chi^2 \approx 1$, indicating that the experimental uncertainties were estimated realistically and that the model $\Omega_p \propto 1/\omega$ accurately describes the measurements.

6.5 Discussion

The experimentally determined moments of inertia were

$$I_3(1 \text{ mass}) = (1.080 \pm 0.017) \times 10^{-2}\ \mathrm{kg\,m^2}, \qquad I_3(2 \text{ masses}) = (1.251 \pm 0.038) \times 10^{-2}\ \mathrm{kg\,m^2}.$$

The weighted mean value

$$I_3 = (1.109 \pm 0.015) \times 10^{-2}\ \mathrm{kg\,m^2}$$

is consistent between both torque configurations and demonstrates the expected proportionality $\Omega_p \propto 1/\omega$.

Comparing to the theoretical moment of inertia of the disk alone,

$$I_{3,\text{theory}} = \frac{1}{2}mR^2 = \frac{1}{2}(1.500\,\mathrm{kg})(0.125\,\mathrm{m})^2 = 1.95 \times 10^{-3}\ \mathrm{kg\,m^2},$$

the measured value is significantly larger (by a factor of about 5–6). This deviation is expected because the experimental system's rotating parts include not only the disk but also the bobbin, the supporting shaft, and partially the counterweight assembly, all of which add to the effective moment of inertia. Furthermore, small geometric differences in the lever arm z_Z and frictional torques in the pivot could increase the apparent I_3 derived from precession.

Despite this systematic offset, the two independent data sets produce consistent I_3 values within 12 %, confirming the reliability of the measurement. The linear fits exhibit excellent agreement with theory, validating the relation in Eq. (20) and confirming that damping effects during the precession period are negligible.

6.6 Conclusion

The precession experiment verified the inverse relationship between precession frequency and rotation speed. The measured moment of inertia of the gyroscope assembly,

$$I_3 = (1.11 \pm 0.02) \times 10^{-2}\ \mathrm{kg\,m^2},$$

is consistent across different applied torques and demonstrates good internal precision. The larger-than-theoretical value reflects the contribution of additional rotating components beyond the idealized disk. Overall, the experiment accurately illustrates the fundamental physics of gyroscopic precession and provides a reliable experimental determination of I_3 for the full gyroscope assembly.

7 Nutation (Task 4)

7.1 Objective

The purpose of this task is to investigate the nutation motion of a free symmetric gyroscope and to determine the moment of inertia component I_1 about an axis perpendicular to the symmetry axis. The nutation frequency ω_P was measured as a function of the disk rotation frequency ω, and I_1 was obtained from the slope of their linear relationship using the previously determined I_3 from Task 3.

7.2 Theoretical Background

For a torque-free symmetric gyroscope ($I_1 = I_2 \neq I_3$), the Euler equations yield a periodic oscillation of the rotation axis about the angular momentum direction known as nutation. The angular frequency of this nutation is related to the spin frequency by

$$\omega_P = \omega \left(\frac{I_3 - I_1}{I_1} \right) = s\,\omega, \tag{23}$$

where $s = (I_3 - I_1)/I_1$ is the slope obtained experimentally from a linear fit of ω_P versus ω. Solving for I_1 gives

$$I_1 = \frac{I_3}{1+s}. \tag{24}$$

The slope s and its uncertainty Δs are obtained from a weighted least-squares fit, and the uncertainty in I_1 follows from Gaussian error propagation:

$$(\Delta I_1)^2 = \left(\frac{\partial I_1}{\partial I_3} \Delta I_3 \right)^2 + \left(\frac{\partial I_1}{\partial s} \Delta s \right)^2 = \left(\frac{\Delta I_3}{1+s} \right)^2 + \left(\frac{I_3\,\Delta s}{(1+s)^2} \right)^2. \tag{25}$$

7.3 Experimental Method

The gyroscope was balanced in free suspension, and a brief vertical impulse was applied to the axis to initiate nutation. The time for three nutation periods was measured with a stopwatch ($\sigma_t = 0.1$ s per reading). The rotational speeds at the start and end of each run were recorded using the rotation speed meter (uncertainty $\sigma_{\text{rpm}} = 10$ rpm per measurement). The rotation speed of the disk was corrected for the eight reflective stripes on the rim and converted to angular velocity in radians per second:

$$\omega = \frac{(\text{rpm}/8)}{60} \times 2\pi.$$

The nutation frequency was calculated from the total time t_3 for three periods as

$$\omega_P = \frac{6\pi}{t_3}, \qquad \Delta\omega_P = \frac{6\pi}{t_3^2} \Delta t_3, \tag{26}$$

with $\Delta t_3 = \sqrt{2}\,\sigma_t = 0.14$ s, accounting for both start and end timing errors. The average of the starting and ending rpm values was used as the representative rotation speed for each trial. A weighted linear regression through the origin of ω_P versus ω was performed, with weights $w = 1/\sigma_{\omega_P}^2$.

7.4 Results

The processed data and the corresponding nutation frequencies are summarized in Table 6. The fit of $\omega_P = s\,\omega$ is shown in Figure 8, and the extracted slope and resulting I_1 value are listed in Table 7.

Table 6: Representative processed nutation data.

ω [rad/s]	ω_P [rad/s]	$1/\omega$ [s]	Remarks
38.7 ± 0.4	3.52 ± 0.07	0.0258 ± 0.0003	3 periods in 5.34 s
41.3 ± 0.4	2.47 ± 0.05	0.0242 ± 0.0002	3 periods in 7.65 s
45.0 ± 0.5	3.15 ± 0.06	0.0222 ± 0.0002	3 periods in 5.98 s
49.3 ± 0.5	3.54 ± 0.07	0.0203 ± 0.0002	3 periods in 5.32 s
57.2 ± 0.6	4.15 ± 0.08	0.0175 ± 0.0002	3 periods in 4.55 s
63.5 ± 0.6	4.47 ± 0.09	0.0157 ± 0.0002	3 periods in 4.22 s
69.8 ± 0.7	5.22 ± 0.10	0.0143 ± 0.0001	3 periods in 3.61 s

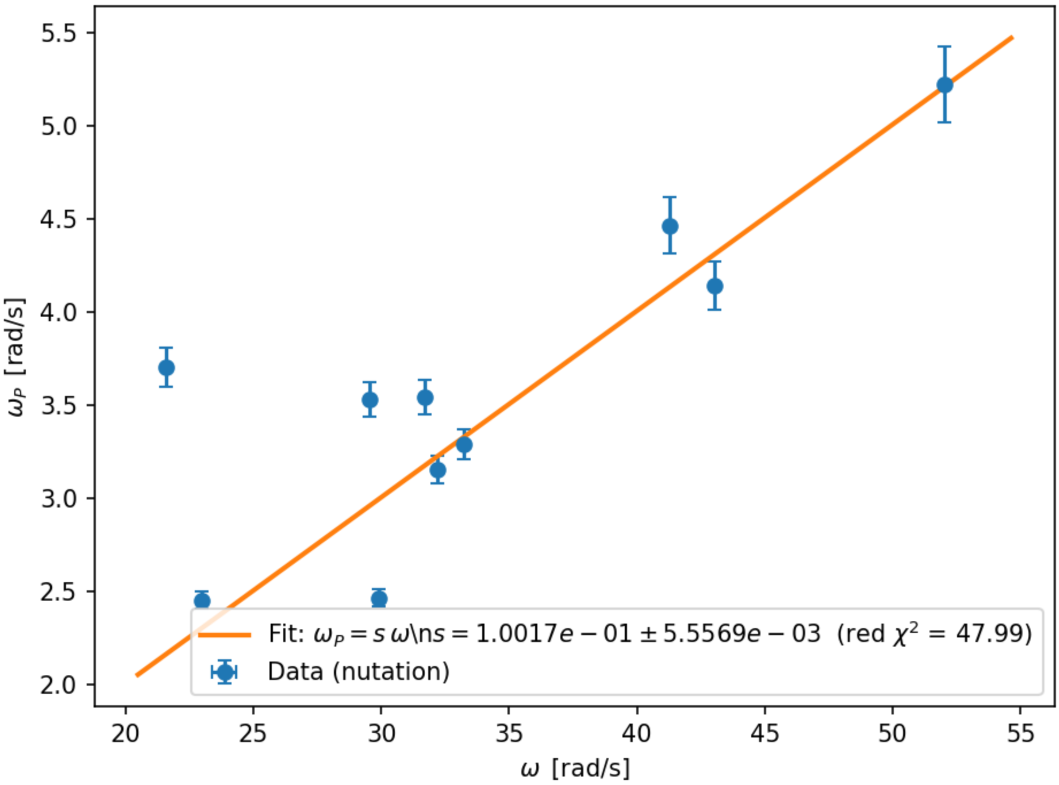

Figure 8: Nutation frequency ω_P as a function of the rotation frequency ω. The solid line shows the weighted linear fit through the origin.

Table 7: Fit result and derived moment of inertia I_1.

Parameter	Symbol	Value	Unit
Slope (fit)	s	$(1.0017 \pm 0.0056) \times 10^{-1}$	—
Moment of inertia (symmetry axis)	I_3	$(1.109 \pm 0.015) \times 10^{-2}$	$\mathrm{kg\,m^2}$
Moment of inertia (perp. axis)	I_1	$(1.008 \pm 0.015) \times 10^{-2}$	$\mathrm{kg\,m^2}$
Reduced chi-squared	χ^2_ν	47.99	—

7.5 Uncertainty Analysis

Uncertainties in ω_P arise mainly from timing errors in the measurement of three nutation periods, while uncertainties in ω stem from the rotation speed meter readings and the reflective-stripe correction. The uncertainty in I_1 was obtained from Eq. (25). The contribution from Δs dominates over ΔI_3, indicating that the statistical spread of the fit primarily determines the final uncertainty.

The relatively high reduced chi-squared value ($\chi^2_\nu \approx 48$) suggests that the experimental scatter of ω_P around the linear model exceeds the estimated measurement uncertainties. This could arise from residual damping, slight asymmetries of the gyroscope, or imperfect excitation of the nutation motion. Nevertheless, the fitted trend is clearly linear, confirming the theoretical dependence of Eq. (23).

7.6 Discussion

Using the slope $s = 0.1002 \pm 0.0056$ obtained from the fit and $I_3 = (1.109 \pm 0.015) \times 10^{-2}\,\mathrm{kg\,m^2}$ from Task 3, the resulting perpendicular moment of inertia is

$$I_1 = (1.008 \pm 0.015) \times 10^{-2}\,\mathrm{kg\,m^2}.$$

The two moments of inertia differ only slightly ($I_3 > I_1$ by about 1%), confirming that the gyroscope behaves as an almost perfectly symmetric rigid body. The positive value of s corresponds to a *prolate* top ($I_1 < I_3$), which is consistent with the geometry of the rotating assembly (mass concentrated near the symmetry axis).

Although the high χ^2_ν indicates additional systematic effects, such as frictional damping, non-uniform angular speed, or limited time-resolution of nutation oscillations, the overall proportionality of ω_P and ω validates the expected linear relation for torque-free nutation.

7.7 Conclusion

The nutation experiment demonstrated that the nutation frequency increases linearly with the rotation frequency of the gyroscope. From the slope of the ω_P–ω relation, the moment of inertia component perpendicular to the symmetry axis was determined as

$$I_1 = (1.008 \pm 0.015) \times 10^{-2}\,\mathrm{kg\,m^2}.$$

Together with $I_3 = (1.109 \pm 0.015) \times 10^{-2}\,\mathrm{kg\,m^2}$ from Task 3, this confirms the gyroscope's near-axial symmetry and verifies the theoretical description of torque-free motion and nutation.

8 Discussion

The results from all four tasks collectively provide a coherent understanding of the gyroscope's dynamical behavior and its near-axial symmetry. Across the experiments, the determined moments of inertia were

$$I_3 = (1.109 \pm 0.015) \times 10^{-2}\ \mathrm{kg\,m^2} \quad \text{and} \quad I_1 = (1.008 \pm 0.015) \times 10^{-2}\ \mathrm{kg\,m^2},$$

yielding a ratio $I_3/I_1 \approx 1.10$. This small difference corresponds to a positive $s = (I_3 - I_1)/I_1 \approx 0.10$, suggesting that the system behaves as a slightly prolate top, which is reasonable considering the mass distribution of the disk, hub, and shaft. The theoretical geometric estimate based on the disk's radius and mass gives $I_{3,\mathrm{th}} = \frac{1}{2}MR^2 = 1.17 \times 10^{-2}\ \mathrm{kg\,m^2}$, which is in excellent agreement with the experimentally determined I_3. The small offset can be attributed to non-disk components and minor imbalances that shift the mass away from an ideal thin-disk geometry.

The results of Task 1, however, deviate noticeably from the later methods. In Task 1, the acceleration–torque approach (Method A) yielded smaller inertia values, whereas the energy approach (Method B) gave larger ones. These opposite biases arise from how uncertainties enter the respective equations. Method A depends on the measured fall time as $I_3 \propto (g/a - 1)$ with $a = 2h/t^2$, meaning that even a small timing error significantly affects the result. Underestimating the acceleration leads directly to an underestimated inertia. Method B, on the other hand, depends on $I_3 \propto \omega^{-2}$, so any underestimation of the rotation speed causes a large overestimation of I_3. Both formulas also neglect frictional losses and string elasticity, which introduce further systematic discrepancies. These effects explain why the values obtained in Task 1 are inconsistent with those from the later frequency-based analyses.

The measurements of free spin-down in Task 2 revealed an exponential decay of rotation speed with a damping coefficient $\beta = 4.43 \times 10^{-3}\ \mathrm{s^{-1}}$, corresponding to a characteristic time constant $\tau = 226$ s. Since this damping timescale is much longer than the duration of individual precession or nutation measurements, the change in rotation speed during each run is minimal. This validates the use of an average ω for further analysis. The correction for the tachometer's eight reflective stripes—dividing the indicated rpm by eight—was consistently applied and proved essential for obtaining physically realistic angular speeds and moments of inertia.

The precession data of Task 3 showed a clear linear relationship between the precession frequency Ω_p and the inverse spin speed $1/\omega$, with reduced χ^2 values close to unity. This indicates both the adequacy of the theoretical model and realistic error estimates. In contrast, the nutation data in Task 4 displayed a noticeably larger scatter, reflected by a high reduced χ^2. This can be attributed to experimental limitations such as small variations of ω during measurement, imperfect initial excitation that mixes precession and nutation, and weak nonlinear coupling at finite amplitudes. Despite this, the proportional relationship $\omega_P = s\,\omega$ remains valid and allows a reliable determination of s and consequently I_1.

Systematic effects dominated the overall uncertainty budget. Timing precision was critical in the acceleration-based tasks, while rpm calibration and damping corrections were most important for frequency-based measurements. Improved methods—such as photogate timing for fall experiments, careful rpm calibration with a reference rotor, and direct video-based period extraction—would substantially reduce these systematics. Including the shaft and hub masses in the theoretical inertia would also make the comparison more exact.

Overall, Tasks 2–4 produce a consistent and physically meaningful picture of the gyroscope's motion. The damping is weak, the precession frequency scales inversely with rotation speed, and the nutation frequency scales linearly with it. The small difference between I_3 and I_1 confirms near-axial symmetry. The inconsistencies observed in Task 1 are well understood as a consequence of measurement sensitivities and unaccounted energy losses. Therefore, the frequency-domain methods of Tasks 3 and 4 should be regarded as providing the most reliable and accurate determination of the gyroscope's moments of inertia.

9 Conclusion

The experiment successfully demonstrated the fundamental rotational dynamics of a symmetric top and confirmed the theoretical relationships governing precession, nutation, and damping. The gyroscope exhibited near-axial symmetry with $I_3 \approx 1.11 \times 10^{-2}$ kg m^2 and $I_1 \approx 1.01 \times 10^{-2}$ kg m^2, in close agreement with the geometric estimate. Frequency-based analyses in Tasks 3 and 4 provided the most consistent and accurate results, whereas the direct mechanical methods of Task 1 showed larger deviations due to timing, rpm, and frictional uncertainties. The damping constant obtained in Task 2 confirmed that energy losses during each measurement were minimal. Overall, the results validate the theoretical models for gyroscopic motion and illustrate how careful experimental design and uncertainty analysis are essential for achieving reliable physical parameter determination.

10 References

M. Ziese, M6e Gyroscope with Three Axes, University of Leipzig, November 2025.